

BANGOR, ME, USA

WMO: 726070

Lat: 44.798N

Lon: 68.819W

Elev: 45

StdP: 100.78

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14606

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-18.7	-27.4	0.3	-20.1	-24.6	0.4	-17.5	12.1	-4.1	10.8	-5.5	2.2	310	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	30.8	21.7	28.9	20.7	27.3	19.5	22.9	28.4	21.9	26.9	20.9	25.2	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	15.9	25.4	20.2	14.9	24.1	19.3	14.1	23.2	67.6	28.4	64.1	27.0	60.6	25.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	8.8	8.0	-26.3	33.8	3.4	1.3	-28.7	34.7	-30.7	35.4	-32.6	36.1	-35.1	37.1
			-26.5	24.8	3.3	1.0	-28.9	25.5	-30.8	26.1	-32.7	26.6	-35.2	27.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-7.4	-5.9	-0.8	6.0	12.4	17.5	20.9	20.0	15.7	9.2	2.8	-3.4
	DBStd	10.75	6.51	5.90	5.28	3.91	3.61	3.55	2.76	2.74	3.84	4.06	4.75	5.58
	HDD10.0	2167	541	446	336	129	14	1	0	0	2	65	219	414
	HDD18.3	4223	799	679	593	369	187	57	8	13	96	284	464	672
	CDD10.0	1194	0	0	2	10	90	226	338	311	174	39	4	0
	CDD18.3	207	0	0	0	0	4	32	87	66	18	1	0	0
	CDH23.3	1763	0	0	3	4	77	314	702	513	144	6	0	0
	CDH26.7	479	0	0	1	1	20	89	198	132	39	0	0	0
Wind	WSAvg	3.4	3.4	3.6	4.1	4.1	3.6	3.2	2.9	2.8	3.0	3.3	3.4	3.3
Precipitation	PrecAvg	1000	75	60	88	86	86	86	80	75	87	90	95	91
	PrecMax	1657	194	133	169	280	194	178	184	167	175	228	295	234
	PrecMin	589	13	14	16	16	14	26	23	18	28	25	26	33
	PrecStd	210	47	27	34	52	49	38	42	39	41	45	51	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	10.0	17.5	22.6	29.4	32.3	32.5	32.5	31.1	24.0	17.1	12.5
		MCWB	10.3	7.7	10.6	13.0	18.4	22.0	22.6	22.3	22.0	16.6	14.1	11.1
	2%	DB	7.5	7.1	11.3	18.1	25.3	29.2	30.7	29.9	27.1	20.2	14.8	9.4
		MCWB	6.0	4.8	6.4	10.3	16.1	20.4	21.6	21.4	19.6	15.0	12.4	8.0
	5%	DB	4.4	4.7	8.7	15.6	22.4	26.9	28.9	27.9	24.5	18.0	12.8	7.2
		MCWB	2.8	2.6	5.0	8.8	14.3	18.8	20.9	20.2	18.3	14.2	10.3	5.4
	10%	DB	2.0	2.8	6.3	13.0	19.9	24.7	27.3	26.2	22.4	16.1	10.6	4.5
		MCWB	0.5	1.0	3.5	7.5	13.1	17.5	19.9	19.1	17.1	13.0	8.6	2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	7.9	11.7	14.9	20.0	23.5	24.6	23.9	22.7	18.6	15.3	11.4
		MCDB	10.6	8.8	15.3	19.6	26.4	30.0	30.1	29.7	29.1	20.4	16.6	12.3
	2%	WB	6.6	5.3	8.0	12.4	17.5	21.7	23.0	22.6	21.1	16.7	13.1	8.5
		MCDB	7.3	6.7	10.0	15.7	23.6	27.4	28.1	27.7	25.2	18.7	14.3	9.2
	5%	WB	2.9	3.0	5.5	10.3	15.7	20.3	21.9	21.5	19.8	15.1	10.9	5.7
		MCDB	4.1	4.3	7.9	13.5	20.5	25.0	27.0	25.9	22.6	17.7	12.1	6.9
	10%	WB	0.6	1.3	3.6	8.6	14.1	18.9	20.9	20.5	18.4	13.3	8.6	3.0
		MCDB	1.7	2.7	5.9	12.4	18.3	22.9	25.4	24.4	21.0	15.6	10.4	4.3
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.6	9.8	10.7	11.4	11.2	11.1	11.5	11.3	10.2	8.7	8.9
		MCDBR	11.4	10.7	12.8	15.7	16.3	14.7	13.2	13.5	13.5	12.7	11.1	10.9
		MCWBR	11.0	9.0	8.8	9.2	8.5	7.4	6.0	6.2	7.2	8.3	9.2	10.0
	5% WB	MCDBR	11.6	9.8	11.3	12.8	13.6	12.7	11.4	11.4	11.2	10.9	9.6	10.7
		MCWBR	11.4	8.9	8.7	8.9	8.0	7.2	5.9	5.9	6.8	8.4	9.0	10.6
Clear-Sky Solar Irradiance	taub	0.278	0.289	0.316	0.347	0.371	0.411	0.425	0.400	0.362	0.327	0.307	0.286	
	taud	2.453	2.408	2.450	2.424	2.397	2.286	2.265	2.331	2.424	2.531	2.551	2.502	
	Ebn at Noon	851	904	922	916	898	859	842	851	859	851	813	806	
	Edh at Noon	70	88	97	109	115	130	131	118	99	78	64	61	
All-Sky Solar Radiation	RadAvg	1.59	2.46	3.66	4.52	5.09	5.52	5.71	5.11	4.14	2.60	1.61	1.22	
	RadStd	0.15	0.22	0.23	0.48	0.63	0.44	0.45	0.30	0.26	0.24	0.15	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+52	N/A

Nomenclature: See separate page